Thanaphumi Kunuthai

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EDUCATION

University of British Columbia Expected Graduation, May 2029

Bachelor of Applied Science (Engineering) *Vancouver, BC*

- *Outstanding International Student Award*

EXPERIENCE

UBC Bionics **Vancouver, BC ,Canada**

Analytics (Machine Learning) Feb 2026 – Present

- Develop and train machine learning models using Python to classify electroencephalogram (EEG) brain signals

ACCESS Venture EWB UBC **Vancouver, BC ,Canada**

Mechanical Engineer Jan 2026 – Present

- Design and prototype an ergonomic, low-cost urine collection device to facilitate non-invasive cervical cancer screening for women in under-resourced African communities.

ACT Technology Club (ACTEC) **Bangkok, Thailand**

Co-Founder & Head of Mechanic August 2022 – March 2025

- Led weekly tech workshops covering AI, data science, programming, robotics, and CAD.
- Mentored award-winning teams to global recognition (Thailand Innovation Awards Finalist, JDIE 2025 Honourable Mention).
- Built and launched a 30km CubeSat with atmospheric sensors to collect predictive weather data.

PROJECTS

Pipe Lie – MLH Production Engineering Hackathon **April 2026**

- **Secured 2nd place overall** and **the Reliability category win** by leading the project's system stability and continuous performance monitoring.
- Engineered comprehensive load tests, identify performance bottlenecks, and ensure high availability.
- Configured real-time Grafana dashboards to visualize core system health metrics and track live performance data.

Explainable Early Detection of Diabetic foot ulcers Device **April 2023 - August 2024**

- **Presented research poster** at the Rice Ken Kennedy Institute **– AI in Health Conference 2024,** Rice University.
- Fine-tuned Vision Transformer (ViT-B/16) and Data-Efficient Image Transformer (DeiT-B Distilled) models on thermal imaging data using class-weighted loss and stratified 5-fold cross-validation to handle dataset imbalance.
- Achieved **91.9% ± 4.9% accuracy / 94.5% ± 3.3% F1** with ViT and **92.2% ± 3.9% accuracy / 94.6% ± 2.8% F1** with DeiT, with ViT showing higher sensitivity **(95.1% ± 4.7% recall).**
- **Awarded Silver Medal at Genius Olympiad 2024 (international)** in Rochester, NY, **and** selected as a **Finalist for the Thailand Innovation Award 2024 ; ranked Top 6 out of 410 teams** in the English pitching competition category.
- My major role is designing hardware architecture by interfacing a Raspberry Pi with a thermal sensor and LCD display, enabling real-time visualization of Grad-CAM heatmaps for on-device screening.

Hydrosense - Hair-Based Absorption System **September 2023 - November 2024**

- Developed a bio-based filtration system utilizing chemically treated human hair to modify keratin structures, achieving removal rates of **81%** for Lead and **63%** for Copper validated by SEM and UV-Vis spectroscopy.
- Integrated **Arduino** and **TDS sensors** to **automate filtration cycles** and facilitate real-time water quality monitoring.
- **Published research** in the **6th KU-IC 2024** and **IJSSHR**, and secured **1st Place** Popular Vote at **I-New Gen 2024** and a Bronze Medal at the NIA STEAM4INNOVATOR Day.

Robotic Dog Medical Assistant **January 2024**

- Developed a robotic assistant prototype by designing **3D models**, **circuit schematics**, and **sensor layouts**
- **Awarded 1st Place** among 66 competitors by the School of Biomedical Engineering at Mahidol University

VOLUNTEERING

King Chulalongkorn Memorial Hospital **August 2024**

- Volunteered at the Elderly Society Clinic to assist staff with measuring patient vitals (BMI, blood pressure) and guided elderly patients in using the hospital's digital appointment application to improve service flow.

Makers Making Change (UBC SBME) **December 2025**

- Created the cost-effective, accessible, switch-activated devices for children with disabilities.

SKILLS

Languages: Python, C, C++, C#, R, Dart, Arduino.
Libraries & Frameworks: PyTorch, TensorFlow, YOLO, NumPy, Pandas, Hugging Face, Flutter, Vision Transformers (ViT).
Developer Tools: Google Cloud, Fusion360, Unreal Engine, Raspberry Pi, Micro:bits, CIRA CORE.